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How Does the Mind Render Streaming Experience as Events?

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Abstract

Events—the experiences we think we are having and recall having had—are constructed; they are not what actually occurs. What occurs is ongoing dynamic, multidimensional, sensory flow, which is somehow transformed via psychological processes into structured, describable, memorable units of experience. But what is the nature of the redescription processes that fluently render dynamic sensory streams as event representations? How do such processes cope with the ubiquitous novelty and variability that characterize sensory experience? How are event-rendering skills acquired and how do event representations change with development? This review considers emerging answers to these questions, beginning with evidence that an implicit tendency to monitor predictability structure via statistical learning is key to event rendering. That is, one way that the experience of bounded events (e.g., actions within behavior, words within speech) arises is with the detection of “troughs” in sensory predictability. Interestingly, such troughs in predictability are often predictable; these regions of predictable-unpredictability provide articulation points to demarcate one event from another in representations derived from the actual streaming information. In our information-optimization account, a fluent event-processor predicts such troughs and selectively attends to them—while suppressing attention to other regions—as sensory streams unfold. In this way, usage of attentional resources is optimized for efficient sampling of the most relevant, information-rich portions of the unfolding flow of sensation. Such findings point to the development of event-processing fluency—whether in action, language, or other domains—depending crucially on rapid and continual cognitive reorganization. As knowledge of predictability grows, attention is adaptively redeployed. Accordingly, event experiences undergo continuous alteration.

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1. Introduction

We exist within a continuously unfolding storm of sensory data. Strikingly, though, this “datastorm” is not what we subjectively experience. Events are what we experience. Events are happenings: units of experience that each singly, and together collectively, culminate in things. We somehow derive such event units from the actual flux of dynamic sensory data; these event units demarcate what happens, help us to encode and retrieve information in memory, support inferences about what will happen next, inform our actions and reactions, and fit our understanding of the causal origins and outcomes of happenings in the world. But how does this work? How do we find, or construct, these causally relevant event units given the fast and furious storm of sensory information we are always confronting? Cognitive science has offered only very partial insight into the mechanisms that underpin this ongoing feat of redescription from sensation to event representations, an achievement that appears to take place largely without conscious awareness.

A recent upsurge in research investigating event redescription processes begins to yield answers to a few basic questions in this arena. In the following review we focus on findings emerging in our own and others’ research regarding three particular questions: What perceptual/cognitive mechanisms underlie fluency in rendering the actual dynamic flow of sensory information as event units? How do these mechanisms operate in the context of novelty for the purpose of learning? What is the developmental story with respect to these event redescription mechanisms?

2. Mechanisms underlying event redescription

Our overarching question, “How does the mind succeed in discerning events as they unfold?,” strikes many as an odd query. The untutored lay perspective seems to be that events are somehow directly experienced. From this perspective, one simply apprehends what is unfolding in the world; no processing of any kind is needed. This “direct perception” intuition leads to the expectation that one’s experience of events will generally be shared by others who were present, awake, and observing things in the same place at the same time. Perhaps for this reason, people tend to be quite surprised by demonstrations showcasing marked diversity in different individuals’ experience of one and the same behavior stream. One demonstration of such diversity is Simon and Chabris’s (1999) famous “inattentional-blindness” study, in which many observers failed to note a very salient event (an actor dressed in a gorilla costume conspicuously passing through) that seems inescapable to others. The eyewitness testimony literature is laden with other

examples; the common thread across examples within these literatures is that viewers report quite different event experiences despite having viewed the self-same sensory stream (e.g., Haber & Haber, 2000; Loftus, 1980; Loftus & Hoffman, 1989). Again, the fact that such demonstrations tend to engender surprise speaks to the power of the everyday intuition that the mind simply apprehends events that veridically occur. Intuition seems to radically underestimate the extent to which event representations are constructed.

The lay intuition that the mind simply plucks events directly from what unfolds in the world also may have hampered cognitive scientists from recognizing event redescription as a worthy topic of study. As it happened, social psychologists initiated both the first explicit consideration of these issues (e.g., Asch, 1952; Heider, 1958) and the first direct empirical work on the topic (Newtson, 1973; Newtson & Engquist, 1976; Newtson, Engquist, & Bois, 1977). In particular, Newtson developed a method for probing adults' on-line experience of events: He recorded the temporal regions they noted when asked to identify meaningful junctures within continuously unfolding activity streams. Briefly, he discovered considerable consistency across viewers in what he called "breakpoint" judgments, found that such judgments reflect nested, hierarchical structure (fine-grained events nested within coarse-grain events), and demonstrated that breakpoint (aka boundary) regions within activity streams coincide with increases in physical motion change and carry higher interpretive value than other (within-event) regions.

2.1. Cognitive and behavioral significance

Since that time, a now sizable body of research has both replicated these findings and extended them in a range of informative ways (see Clewett & Davachi, 2017; Kurby & Zacks, 2008; Levine, Buchsbaum, Hirsh-Pasek, & Golinkoff, 2018; Richmond & Zacks, 2017 for recent reviews). Perhaps most importantly, this literature underscores that observers' fluency in identifying boundaries as activity unfolds holds crucial significance for their more general cognitive functioning. Along these lines, highlighting boundary regions facilitates viewers' subsequent memory for complex activity sequences (Gold, Zacks, & Flores, 2017; McGatlin, Newberry, & Bailey, 2018), and boundary identification predicts ability to enact such sequences (Bailey, Kurby, Giovannetti, & Zacks, 2013). Converging evidence from fMRI reveals that the strength of hippocampal activity specifically at event boundaries is predictive of later recall (Baldassano et al., 2017). Event boundaries seem to elicit specific patterns of neural activation in non-human species (e.g., rodents and non-human primates) as well, with systematic links to behavioral functioning (Bilkey & Jensen, 2019). Furthermore, human adults who display disruption in detecting boundaries during event viewing display reduced event memory (Kurby & Zacks, 2011; Sargent et al., 2013) and are at risk for cognitive impairment (e.g., Bailey et al., 2013; Zacks, Kurby, Landazabal, Krueger, & Grafman, 2016; Zacks, Speer, Vettel, & Jacoby, 2006). Apparently, when fluency in detecting event boundaries is impaired, experience loses its crispness and encodability, thereby undercutting comprehension, memory, and re-enactment.

2.2. *Why privilege boundaries?*

Together, this growing literature makes clear that fluent observers privilege boundary regions in their processing of streaming experience. But why might this be? An additional body of work offers beginning ideas about why, and points to some of the important mechanisms that may be operating in viewers' identification of boundaries as sensory information streams past. We briefly review such evidence, and we note that Shin and Dubrow (2020) also focus on reviewing existing research relevant to answering this question. An influential idea that emerged from the initial findings is that event perception may often ride on viewers' expectations or predictions about what is to come. In other words, there is predictability to be found within the stream of sensation one experiences (and creates through one's own activities); perception captures that structure and capitalizes on it to project future experience, plan for it, and coordinate (or successfully compete) with others to optimize functional success (Baldwin & Baird, 1999, 2001; Zacks & Tversky, 2001; Zacks, Tversky, & Iyer, 2001; but see Schapiro, Rogers, Cordova, Turk-Browne, & Botvinick, 2013 for evidence of alternate mechanisms for event segmentation). Baldwin and colleagues suggested that viewers (whether adults or infants) may achieve an analysis of the boundary-related structure within continuous activity in part by detecting predictability structure; that is, statistical regularities. In particular, they suggested that movement sequences with high transitional probabilities would come to be experienced as cohesive units, whereas regions of low transitional probability would emerge as boundaries between such event units (Baldwin, 2002; Baldwin & Baird, 2001).

2.3. *Event Segmentation Theory*

In 2007, Zacks and colleagues proposed a highly influential account—Event Segmentation Theory (EST; Radvansky & Zacks, 2011; Zacks, Speer, Swallow, Braver, & Reynolds, 2007)—offering an elaborated and more explicit formulation of these ideas. They emphasized that event units arise within unfolding experience when knowledge of predictability from prior experience yields prediction *failure*; events gain the sense of being bounded when a mismatch occurs between one's prediction efforts and what actually occurs. In their account, it is when transient prediction error—"surprisal"—spikes that one experiences the culmination of an event, just as it transitions into the next. If this is correct, event units represent packages or chunks of high-predictability activity that are bounded by the unexpected.

But why would EST posit that event boundaries tend to occur at regions of transient prediction error? The idea here is that once actors set a goal in motion, much is likely to be predictable about what they do (once one launches a reach for a cup, for example, things tend to unfold predictably with respect to the progression of that reach). At a certain point, however—often just when a goal is achieved—predictability plummets. To pursue the earlier example, once the cup is grasped, any number of other activities could ensue—lifting the cup, pushing it, holding it, spinning it, and so on. And yet, once the

next goal-oriented activity is clearly underway, predictability once again improves considerably; that is, until the next boundary occurs.

Thus far, we have characterized EST's operation in relation to observers achieving an analysis of human-generated, intentional activity. It is important to note that EST (as well as extensions to EST we propose here) provides a unified account of viewers' processing of sequences more generally, regardless of whether those sequences involve goal-oriented human action or arise from physical mechanisms that operate on objects. That is, many event sequences occur outside of human agency; imagine a feather buffeted by the wind ultimately settling on a decaying log, or a boulder breaking loose, hitting another with cascading consequences that ultimately yield a massive avalanche. Such sequences also have predictability structure (arising from physical and chemical principles that constrain the behavior of all things) that the human mind apprehends. Identification of event boundaries within such physical sequences can occur via the same prediction-failure mechanism by which EST accounts for segmentation of human activity sequences.

Event Segmentation Theory (EST) is consistent with phenomena already described, as well as a growing body of newer findings. Among other things, Zacks, Kurby, Eisenberg, and Haroutunian (2011) found that, as EST predicts, viewers report heightened uncertainty at boundaries regarding what will happen next, relative to other regions within unfolding activity. As well, looking behavior seems to reflect reduced predictability at boundaries, in that viewers display earlier predictive looking to objects contacted mid-stream within events relative to objects contacted at event boundaries (Eisenberg, Zacks, & Flores, 2018). Moreover, Reynolds, Zacks, and Braver (2007) developed a computational model implementing the basic tenets of EST that reliably detects event units within streaming human activity via predictability monitoring.

Among other benefits, EST offers an explicit formulation of event processing that provides at least an initial account for how the mind might cope with the ubiquitous degree of sensory novelty inherent in everyday experience. We turn now to focus more specifically on this issue. To foreshadow, recent work on this topic yields an important theoretical reformulation of the mechanisms involved.

3. Learning: Discovering events within novelty

It follows straightforwardly from EST that perception of event units should change as the predictability structure within experience changes across time. Thus, a corollary of EST concerns learning: Event boundaries should emerge in viewers' experience as they gain familiarity with the predictability structure within unfolding activity. In confirmation, a number of statistical-learning studies demonstrate that viewers readily discover event units in novel activity sequences when predictability structure is the sole source of information about these newly identified event units. Moreover, as EST would predict, boundaries between these newly discovered event units are precisely those junctures within the continuously unfolding activity stream at which predictability is lowest (e.g., Avrahami & Kareev, 1994; Baldwin, 2002; Baldwin, Andersson,

Saffran, & Meyer, 2008; Buchsbaum, Griffiths, Gopnik, Plunkett, & Baldwin, 2015; Swallow & Zacks, 2008). It is worth noting here that research utilizing directly comparable statistical-learning techniques in the language domain reveals analogous findings: Listeners encountering continuously streaming novel auditory sequences likewise detect units (e.g., word-and/or clause-level units) solely via sensitivity to the predictability structure within those auditory streams (e.g., Astheimer & Sanders, 2011; Hartshorne et al., 2019; Saffran, Aslin, & Newport, 1996). Such comparability across domains hints that the self-same learning mechanisms are at play despite differences in modality and the details of the structural regularities involved. If this is correct, then a clear implication is that models developed to account for event perception encompass language processing and language acquisition as well. Indeed, several theorists have called for just such an overarching, integrative account of mechanisms enabling the mind to delineate structure and meaning within continuously streaming sensory information (Baldwin, 2005; Cassimatis, 2006; Christiansen & Chater, 2016; Frost, Armstrong, Siegelman, & Christiansen, 2015; Kirkham, Slemmer, & Johnson, 2002; Saffran & Kirkham, 2018; Saffran & Thiessen, 2007; Zacks et al., 2007).

3.1. Predictability monitoring as data compression

What is the point of a cognitive system that goes to considerable trouble to track predictability and use it to define event units? Such a system can radically reduce the information-processing challenge—the “datastorm”—that the mind continually confronts. What is predictable can be compressed into short-hand, more “gist-level” representations in memory, thereby relieving the mind of coping with the massive ongoing barrage of sensory detail (e.g., Bransford & Franks, 1971).

On this view, event perception is a form of data compression, which can have great psychological value if—as recent findings clearly confirm—the chunks derived from this compression process are amenable to other cognitive operations, such as memory encoding, memory retrieval, hierarchical organization, linguistic labeling, and causal inference (e.g., Brunec, Moscovitch, & Barense, 2018; Buchsbaum et al., 2015; Chekaf, Cowan, & Mathy, 2016; Christiansen, 2019; Christiansen & Chater, 2016; Miller, 1956; Shin & Dubrow, 2020).

3.2. Information optimization: Modulating attention to optimize processing efficiency

In addition to facilitating cognitive function such as memory, chunking streaming sensory data into high-predictability event units also offers a way to enhance efficiency by reducing the outlay of attentional resources as processing is underway. Processing can be optimized to selectively sample just those regions within unfolding activity that are likely to be highly informative (boundary regions), while suppressing sampling of content already known to be highly predictable, and thus less informative. In other words, viewers unconsciously pulse attention toward information-rich regions within the sensory flow. However, these regions are not information-rich in an absolute sense—they derive their

richness by virtue of the predictability structure of a given viewer's accumulated experience, taken in relation to the viewer's goals at any given moment in time.

Research employing a deceptively simple new methodology developed by Hard, Recchia, and Tversky (2011)—the dwell-time paradigm—elegantly showcases this “information-optimization” strategy viewers spontaneously adopt that enables them to achieve compressed sampling of the sensory stream over the course of their experience. Hard and colleagues asked viewers to advance at their own pace through slideshows depicting an actor undertaking continuously unfolding familiar activity, such as organizing a college dorm-room. Quite unconsciously, viewers tended to dwell longer on slides depicting boundary than within-event content, suggesting a tacit tendency to optimize attentional resources by selectively sampling highly informative (low-predictability) boundary content over less informative (high predictability) within-event content. This finding has since been replicated in several studies with a sizable range of familiar activity sequences (Baldwin & Sage, 2013; Kosie & Baldwin, 2019a, 2019b; Smith, Newberry, & Bailey, 2020).

3.3. Compression-tuning within novel activity streams

Other recent work by Hard and colleagues using the dwell-time paradigm illuminates the developmental emergence of information optimization as viewers progressively gain familiarity with sequences that were initially novel to them. The underlying hypothesis was that, when confronting novel sequences, information-optimization processes enable the mind to rapidly and implicitly tune to predictability structure as it is experientially discovered. To test the hypothesis, Hard, Meyer, and Baldwin (2019) presented adults with novel activity sequences whose higher-level segmental structure was defined solely via statistical regularities. One group of adults (knowledgeable participants) viewed a video of an extended activity corpus embodying the predictability structure of the novel sequence. Subsequently, they were asked to advance at their own pace through a slideshow depicting the identical sequence. A second group (naïve participants) advanced through the same slideshow, but without any prior opportunity to view the activity corpus (thus having lacked the chance to discover the predictability structure of the sequence prior to slideshow viewing). The information-optimization account predicts that knowledgeable viewers' dwell times will display signs of information compression, which should take the form of detectable dwell-time surges at boundaries between statistically defined units. Such marked surges at higher-level boundaries should be largely absent in naïve participants' dwell times. These predictions emerged clearly in participants' dwell-time patterns. Moreover, individual differences in the strength of participants' dwell-time surges at boundaries predicted their later accuracy on a discrimination task measuring the degree to which they had registered the statistically defined segmental structure of the novel activity sequence.

Together, these findings provided the first direct window on the emergence of viewers' compression-tuning to novel predictability as it unfolded within their experience. The findings also confirmed one of the advantages of the information-optimization account

over EST. On EST, it is not clear why knowledgeable viewers—who had had more experience with the novel activity sequence than naïve viewers—would show a relatively greater response to high-level boundaries; in particular, on EST these boundaries ought to have elicited *less* surprisal (and thus garnered less attention) for knowledgeable participants than for naïve participants. In contrast, Hard and colleagues' findings—that viewers' attention reorganizes with increasing familiarity to selectively target high-level boundaries—arise naturally out of the information-optimization account.

The findings just described provide clear-cut evidence that attention is reshaped as learners gain familiarity with predictability structure. This attentional reorganization capitalizes on growing knowledge of predictability structure to increase efficiency in the outlay of attentional resources. In the Hard et al. (2019) research, what learners are learning to predict is well-specified: They are gaining increasing ability to predict which small-motion elements involving an object (e.g., inspect, touch, insert, etc.) predict the next. Our information-optimization account proposes that the new, higher-level event units that learners gain access to in this way yield new chunks of experience over which predictability structure can now be registered. In this way, optimizing progressively yields new inputs to statistical learning and hence ever-new targets for attentional reorganization (see Saffran & Wilson, 2003, for a similar suggestion regarding language learning). In this way, an optimization mechanism that takes input from predictability tracking processes can yield a powerful force for abstraction and perceptual/cognitive change.

3.4. *Optimizing benefits fluency*

In regard to the information-optimization account we are forwarding, one might ask what the cost would be of failing to optimize. We suggest that failure to optimize reduces event-processing fluency. This is because optimizing is necessary if one is to effectively process the barrage of sensory information one is invariably confronted with. But what precisely does event-processing fluency mean? It means the ease and accuracy with which events are comprehended and recalled. A classic example of fluency differences in event processing involves differences between experts versus novices in a given domain. Consider the game of lacrosse, for example. As a lacrosse novice, one frequently fails to catch key junctures in the highly dynamic, rapidly unfolding game that are easy for a lacrosse expert to pick out. As a result, relative to the lacrosse expert, the lacrosse novice has difficulty registering and comprehending what is happening, and thus incomplete, even fragmented recall. As well, a novice may recall what to an expert seem to be irrelevant details. More generally, the novice fails to optimize as a result of not having yet had the relevant real-world experience needed to acquire the predictability structure that supports expert-level optimization. Such novice versus expert differences are reminiscent of comprehension and recall deficits that Zacks et al. (2006) documented in a comparison of event processing in adults with mild dementia versus age-matched unaffected adults. In the case of dementia, however, optimization presumably fails due to disease-related disruptions to neurological mechanisms supporting information optimization.

Tuning processing to capitalize on patterns detected in the sensory stream appears to be a domain-general propensity in humans. This gives rise to further speculation about an integrative account of event perception. In particular, tuning to optimize information uptake within streaming activity has previously been observed in speech perception (Kuhl, 2004; Werker & Tees, 1984), featural processing in face recognition (Kelly et al., 2007; Mondloch et al., 1999), processing of emotional facial expressions (Pollak, Messner, Kistler, & Cohn, 2009), and featural processing in human action processing (e.g., Loucks & Baldwin, 2009; Loucks & Somerville, 2012a).

3.5. *Additional benefits of information optimization*

As a means for demarcating experience, information-optimization boasts real advantages; among these are *resilience* and *flexibility*. Swallow, Kemp, and Simsek (2018) recently discovered that event segmentation differs little when activity is recorded from a first- versus third-person perspective, despite the fact that the visual information associated with these perspectives is markedly different. The information-optimization approach readily accommodates such segmentation stability in the face of differing visual perspectives, in that regions of predictable-unpredictability would be expected to overlap heavily for first- versus third-person perspectives. Another kind of stability that seems to be characteristic of event perception is resilience in the face of missing information. Relevant content within streaming behavior is often absent or degraded, whether for reasons of occlusion, distance, poor lighting, or viewer distractibility. How does processing change in the face of such missing information? Interestingly, reducing the resolution of the sensory stream even by as much as half leaves viewers' pattern of sampling strikingly unaffected (Kosie & Baldwin, 2019b). Of course, such resilience cannot be absolute: At some point the degree of missing information will exceed the viewer's ability to interpret the kinds of activities underway, and thus the ability to engage in prediction will be disrupted. The precise point at which such breakdown occurs is itself an interesting question for future research, and it may depend on a variety of factors. To the extent that resolution is good enough to support prediction, the selective sampling of boundary content characteristic of information optimization seems to be largely unaffected, making processing impressively resilient to missing content.

Perhaps, however, one should consider the possibility that what appears to be resilience in the face of degraded content is simply a result of viewers adopting a rigid sampling pattern that is unresponsive to change. Fortunately, this "rigidity" alternative has been easy to rule out. In recent work, we found that a simple change to viewing instructions alters viewers' dwell-time patterns substantially (Kosie & Baldwin, 2018). For example, viewers asked to watch a novel method of shoelace tying in order to later enact it themselves sampled the activity stream in distinctive ways relative to viewers simply asked to remember the new method. Stated more generally, viewers flexibly reorganized processing in relation to the specific instructions they were given.

Clearly, what is information-rich within a given sensory stream is not fixed, but depends, at least in part, on context and/or viewers' processing goals (see also Clewett &

Davachi, 2017; Smith et al., 2020). Adaptive flexibility in regard to the information viewers target for sampling—and hence the events they register—seems key to successful cognitive functioning for a species such as humans who thrive while operating in diverse and rapidly changing environments.

3.6. *Revising Event Segmentation Theory*

As we alluded to earlier, evidence for information optimization in event perception points to a necessary revision to EST. We note that Shin and Dubrow (2020) also suggest important, but different, revisions to EST.

The standard EST proposal pins boundary detection on “surprisal”; that is, spikes in prediction error (but see Richmond & Zacks, 2017, for consideration of possible “tweaks” to the standard proposal). Yet findings from recent dwell-time studies clearly demonstrate that viewers are not simply reactive to low-predictability regions within streaming behavior; rather, heightened sampling of the activity stream systematically launches in *anticipation* of boundaries (Hard et al., 2011, 2019; Kosie & Baldwin, 2019b). The significance of this is that viewers clearly are predicting boundaries; that is, they are predicting the unpredictable. Thus, simple surprisal—the identification of prediction error—is not all that drives the experience of bounded events; rather, the experience seems also to arise when low-predictability regions are *expected* and selectively targeted for heightened attention.

One might query whether there is indeed a need to interpret surges in dwell time that launch in advance of boundaries as arising from predictions of upcoming unpredictability. Such increases might not seem incompatible with EST, which would account for these surges as occurring in *response* to prediction error-associated changes in perceptual and/or behavioral variables that naturally precede boundaries. However, the learning data we reviewed earlier (e.g., Hard, Meyer, & Baldwin, 2019) provide strong evidence in favor of the optimization account’s anticipation framing of these pre-boundary surges in dwell time. Recall that it is only after extended exposure to a novel sequence that viewers display these anticipatory surges in dwell times. Without such exposure to the identical sequence, these surges are lacking. This finding clarifies that perceptual changes in the motion itself are not solely responsible for the timing of dwell-time surges; rather, these anticipatory increases in dwelling seem to reflect attentional reorganization attendant on knowledge of predictability structure. The optimization account provides a straightforward explanation for this reorganization. Specifically, anticipation of boundaries enables attention to be selectively oriented to temporal regions within activity where uncertainty is predictably highest, thereby optimizing processing by harnessing acquired knowledge of predictability structure.

Put another way, demarcation of events within experience may arise both as a result of what Braver, Gray, and Burgess (2007) term “proactive processes”—in the case of event processing, the implicit sampling of regions high in predicted uncertainty—as well as via “reactive” operations that orient to surprises that the environment dishes out exogenously. In addition to accounting for findings that a surprisal-alone account does not, this two-

fold framework also has the benefit of providing a more integrative account of developmental phenomena, a point we explore in more detail in a later section. We note also that our information-optimization proposal bears interesting resemblance to other theoretical accounts, such as Friston's "free-energy principle" (Friston, 2009), Christiansen and Charter's (2016) characterization of language processing as "chunk and pass" which enables anticipation, and a theory of "interoception" recently forwarded by several research groups (e.g., Barret, 2017; Barrett & Simmons, 2015; Craig, 2002; Seth, 2013; Seth, Suzuki, & Critchley, 2012) to account for processing of complex phenomena such as emotions and emotional facial expressions.

Two studies from our research lab provide additional empirical evidence confirming the involvement of both endogenous (proactive) and exogenous (reactive) influences on predictability monitoring. In these studies, adults' dwell times were recorded as they viewed event sequences specifically designed to violate causal expectations at specified points within the continuously unfolding activity. In one study (Sage & Baldwin, 2014) these were sleight-of-hand sequences, in which an experienced magician produced seemingly impossible occurrences, such as passing apparently solid rings through one another. The other study (Baldwin, 2011) presented adults with matched sequences that involved identical activity streams: in one case a mundane causal event (a hand grasped the stick of a cane, and in the course of pulling, the cane brought a stuffed toy within reach so that the hand could grasp the toy) versus a causally impossible version of the event (the pull of the cane impossibly brought the toy within reach despite the fact that the toy was only near the cane, not within its arc). Across both of these studies involving violation-of-expectation sequences, adults' dwell times replicated the typical "boundary advantage" in viewers' dwell times (i.e., which we interpret as indicative of endogenously driven heightened sampling of information-rich boundary regions). As well, however, dwell times also surged in response to specific junctures at which violation-of-expectation (i.e., surprisal) occurred. That is, when prediction failure *unexpectedly* occurred, dwelling reactively increased markedly. Importantly, this was the case despite the fact that these surprisal junctures did not necessarily coincide with event boundaries (as judged in relation to comparable sequences in which expectations were not violated). These findings clearly point to information optimization involving both a proactive, endogenous process—by which the viewer heightens attention in conjunction with actively anticipating uncertainty—and an exogenous, reactive, surprisal process—by which the viewer responds to unexpected prediction error with increased attention. The proactive and reactive processes we describe here can perhaps best be thought of as two sides of the predictability monitoring coin. Both processes hinge on predictability monitoring, but the proactive process harnesses knowledge of predictability structure to optimize processing of the data, enabling chunking of event units (aka compression). In contrast, the reactive process relies on knowledge of predictability to identify regions within experience that are genuinely unexpected—not yet analyzed or understood—which then can be targeted for further analysis in order to discover the as-yet-opaque predictability structure.

3.7. *Current limitations*

Although our information-optimization account is currently consistent with a broad swath of empirical evidence, it is as yet far from complete. Much remains to be specified. For instance, the account ultimately needs to be able to specify, for any given unfolding sensory stream, (a) what the learner is learning to predict (i.e., for what sensations or abstractions is predictability monitored?), (b) how such prediction works (e.g., what is the precise nature of the event models/schemas/representations that are derived?), and (c) specifically what information is used to adapt the predictions (e.g., what kinds of error signals operate and how are they used to effect change in other processes?). At this early phase, we point to important work by others that makes progress on such issues as (a) the nature of event models and the processes by which such models are updated (e.g., Chen, Lu, Beukers, Baldasson, & Norman, 2019; Hommel, Müsseler, Aschersleben, & Prinz, 2001; Johnson-Laird, 2010; Richmond & Zacks, 2017; Stawarczyk, Bezdek, & Zacks, 2019), and (b) ways in which event predictive cognition interfaces with execution of action (e.g., Belardinelli, Lohmann, Farnè, & Butz, 2018; Lohmann, Belardinelli, & Butz, 2019). It will also be important to discover the extent to which our optimization account meshes versus conflicts with other emerging explanations, such as that of Shin and Dubrow (2020), regarding how viewers register event boundaries as sensory information flows past. Progress on these fronts will be an important direction for further work regarding information optimization.

3.8. *Interim summary*

To sum up thus far, we round back to the overarching issue of event discovery in the face of novelty. As biological entities, we are awash in novelty. Barring modern inventions such as video-replay technologies, sensory experience never repeats itself in full detail. Thus, event perception mechanisms had to have evolved to cope effectively with novelty. A strategy of monitoring and harnessing predictability wherever it can be found is part of such coping. In a sense, this is how the novel recedes into becoming the familiar while at the same time the mind incorporates what is new. In particular, we suggest that the mind harnesses knowledge of predictability to progressively optimize sampling of the sensory flow—selectively targeting regions of predictable-unpredictability (boundaries) and suppressing attention to regions of high predictability. We propose that this implicit, proactive, optimization strategy is one mechanism giving rise to the demarcatedness of experience. That is, as the targeted sampling of information-optimization process plays out, it delivers chunky, “event-populated” experience. This unconscious, predictability-tuned compression process is efficient and robust, yet clearly also flexible. Understanding more about the nature of such flexibility will be an important leading edge for future investigation. Next, we consider what is known regarding the developmental trajectory of event redescription skill.

4. Development: Acquiring fluency in rendering experience as events

There is little doubt that event perception undergoes massive developmental change from infancy through adulthood, for the obvious reasons that (a) knowledge undergoes remarkable expansion with development, and (b) knowledge plays a key role in shaping the nature of events one is even able to discern. A human infant presumably cannot experience some events that seem very mundane to many adults—events such as *cashing a check*, for example—because knowledge of relevant social and cultural systems (e.g., money, financial institutions, paper-based monetary transactions, etc.) is required to even conceive of such an event. However, even though infants almost certainly cannot conceive of check-cashing, per se, they may nevertheless be able to demarcate the flow of behavior into relevant units. That is, infants with adequate visual acuity might be able to discern fine-grain events such as an actor stepping up to a counter, extracting paper from a wallet, placing paper on a counter, passing paper to a person on the far side of a counter (teller), and the like; they might also be able to discern the more coarse-grain level event of an actor having an extended interaction with a person behind a counter. However, one might ask how infants would find such chunks within continuously flowing dynamic activity despite their naïveté. Baldwin and Baird (1999) suggested a bootstrapping solution to this question some years ago; they proposed that initial segments or chunks might emerge in infants' experience as a result of their sensitivity to structural patterns within the motion flow itself that happen to correspond with units inherent to goal-directed activity. Sensitivity to such patterns within the motion flow could give infants initial purchase on potentially relevant chunks within dynamic human action in relation to which they could further monitor predictability and pursue information optimization. This proposal emerged on direct analogy to infants' previously documented sensitivity to intonationally cued units within continuous speech (e.g., Jusczyk et al., 1992).

In concert with these ideas, we propose that a basic propensity to demarcate ongoing sensory experience into unitized chunks (events) via optimization processes is an inherent property of human perception and cognition and thus not acquired. Put another way, we suggest that the mind's event-oriented compression strategy is a pre-adapted and universal aspect of human perceptual/cognitive functioning that is operative from the advent of perceptual/cognitive functioning in human ontogeny. If this is correct, a major driver of cognitive developmental change—and in particular, progress in event-processing fluency—resides in the learning that optimization processes yield as experience accrues. On this view then, optimization processes, which shape and reshape the deployment of attention across time and thereby alter experience, are a primary engine behind learning and perceptual/cognitive change across the lifespan. Put another way, we propose that event redescription processes, being optimization processes, are inherently learning mechanisms that generate cognitive developmental change across time.

We turn now to summarizing evidence regarding event perception in infancy and early childhood. This body of work provides initial grounding for our information-optimization

formulation of cognitive development, and it suggests fruitful directions for future investigation.

4.1. Tell-tale signs of information optimization early in life

Abundant evidence now confirms that event segmentation processes akin to those of adults are operative early in life (see Levine et al., 2018 for a recent review). Research on language development pioneered documentation of such skill on infants' part (see Saffran & Kirkham, 2018 for a recent review), with comparable skills having been documented more recently in the context of processing dynamic human activity streams. These findings all point to infants possessing skill at monitoring predictability structure within unfolding experience and progressively optimizing attention to sample particularly informative regions within unfolding experience. For example, infants not only display sensitivity to event boundaries in everyday intentional action sequences (Baldwin, Baird, Saylor, & Clark, 2001; Hespos, Grossman, & Saylor, 2010; Hespos, Saylor, & Grossman, 2009; Saylor, Baldwin, Baird, & LaBounty, 2007; Pace, Levine, Golinkoff, Carver, & Hirsh-Pasek, 2020); they also readily capitalize on predictability structure to discover new event units in novel activity streams (Baldwin, 2012; Roseberry, Richie, Hirsh-Pasek, Golinkoff, & Shipley, 2011; Stahl, Romberg, Roseberry, Golinkoff, & Hirsh-Pasek, 2014) and display better memory for event content occurring at boundary regions relative to within-event content (Sonne, Kingo, & Krøjgaard, 2016, 2017). Several lines of evidence indicate that infants can register predictability structure within human activity in relation to actors' goals, as opposed to just the specific surface motion patterns actors adopt to pursue those goals (Olofson & Baldwin, 2011; Woodward, 1998). They can also organize their processing of goal-directed activity around that predictability structure by, for example, directing gaze to an actor's predicted goal object in advance of the actor's actual contact with that object (Ambrosini et al., 2013; Kanakogi & Itakura, 2011; Monroy, Gerson, & Hunnius, 2017). Moreover, infants display developmental reorganization in their processing of goal-directed activity that reflects progressive tuning to the predictability structure specific to a given activity type (Loucks & Sommerville, 2012a, 2012b) and capitalize on their own motor experience to guide such tuning (Loucks & Sommerville, 2012b; Sommerville & Woodward, 2005; Sommerville, Woodward, & Needham, 2005). This body of findings—only a small portion of which we have mentioned here—clearly points to the operation of the proactive processes posited by the information-optimization account.

Further evidence also clearly substantiates complementary reactive processes in operation during infancy as well. For example, infants are well known to react with enhanced attention to the unexpected. This is such a well-understood phenomenon of infancy that a great deal of modern knowledge about infant perception and cognition is grounded in measuring (e.g., via looking, sucking, etc.) infants' interest in, and behavioral response to, what is novel or violates expectation. This literature thus provides a wealth of documentation for the operation of predictability-guided reactive processes in infants' event processing.

A large literature involving preschoolers (e.g., Gentner, 2010; Gopnik, 2012; Legare, 2012; Legare, Gelman, & Wellman, 2010; Schulz, 2012) and a sizable collection of infancy studies also document that young children flexibly update and reorganize their processing of events in response to new or unexpected information, pointing to the progressive reorganization of processing that information optimization potentiates. A clever study by Woodward and Sommerville (2000) provides one example: They observed that 12-month-old infants' processing of an ambiguous event stream (an actor touching the lid of a box) differed depending on the causal nature of a larger event it was embedded within; in particular, only infants who previously witnessed the actor remove an object from the box went on to encode the lid-touching act (in isolation) to be directed toward the object inside the box. Clearly, what infants gleaned from the preceding flow of activity informed their analysis of the structure within the subsequent activity stream; they updated their event analysis in relation to relevant new information. An especially striking illustration of information optimization at work reshaping processing is seen in recent studies of Stahl and Feigenson (2015, 2017). They intentionally confronted infants with a violation-of-expectation experience, and observed that this experience triggered infants to upregulate attention to the object central to the violation experience, and subsequently engage in directed exploration of this object of interest. These studies stand as yet another clear indication that infants' processing is organized around their predictions, and violation of these predictions elicits attention and triggers systematic efforts directed toward further analysis. Lastly, Kidd and colleagues (Kidd, Piantadosi, & Aslin, 2012; Piantadosi, Kidd, & Aslin, 2014) documented, both empirically and computationally, that infants as young as 7–8 months of age optimize use of attentional resources in their sampling of unfolding information. Specifically, they found that infants' looking conforms to a "Goldilocks Effect"; infants avoid devoting attention to information that is either overly simple or overly complex. Together, all of these sources of evidence mesh nicely with our proposal: Information-optimization processes seem to be foundational to learning from earliest infancy and subserve continuous, progressive reorganization of processing in ways that produce optimal tuning to discoverable structure within the unfolding sensory flow.

4.2. What is the starting state?

Like any developmental account (e.g., Gopnik, Meltzoff, & Bryant, 1997), ours will not be complete until we can specify the nature of infants' initial processing of the datastream in which they are immersed. After all, innate information-optimization mechanisms can only optimize if there is something to optimize over. As yet, much remains mysterious regarding the nature of infants' starting state in terms of what experiences they extract from the dynamic, multimodal flow of sensation. Unresolved debates flourish (see Elsner & Adam, 2020, for a comprehensive review) regarding the extent to which specific cues within the sensory stream, such as autonomous motion, might trigger infants to construe activity not only as an event, but a specific type of event; namely, an agentive, goal-directed event. Fortunately, intensive investigation in some domains, such as that of

speech perception and face perception, has begun to yield a picture of what initial processing biases might be, and how processing is transformed in relation to the specifics of infants' experience (e.g., Johnson, 2011; Kuhl, 2004; Vouloumanos & Werker, 2007). This impressive body of work provides a wealth of ideas for how to proceed in investigating comparable issues in regard to event processing more generally.

4.3. Information optimization often privileges the social

As many have remarked, social factors seem especially important in shaping infants' registration of predictability structure and hence the path along which processing reorganization proceeds. For example, abundant evidence indicates that infants are more inclined to register the predictability structure of content couched within a highly social format (e.g., Baldwin et al., 1996; Kuhl, 2004; Sage & Baldwin, 2011). An even larger body of evidence indicates that social others—especially caregivers—spontaneously scaffold infants' auditory, visual, and kinesthetic experience in ways that support infants' discovery of predictability structure. For example, infant-directed speech (motherese) and action (motionese) exaggerates internal predictability structure and intention/goal-related information (Brand, Baldwin, & Ashburn, 2002; Brand, Hollenbeck, & Kominsky, 2013; Fernald, 1989; Fernald & Mazzie, 1991; Fernald & Simon, 1984; Kuhl, 2004; Meyer, Hard, Brand, McGarvey, & Baldwin, 2011; Nomikou & Rohlfing, 2011). Infants prefer such predictability-enhanced sensory streams (e.g., Brand & Shallcross, 2008; Cooper & Aslin, 1990; Fernald, 1985; Pegg, Werker, & McLeod, 1992) and often display superior learning in relation to such enhancements (e.g., Brand & Tapscott, 2007; Kuhl, 2004; Matheson, Moore, & Akhtar, 2013; Shneidman, Todd, & Woodward, 2014; Williamson & Brand, 2014; but see Shneidman, Gaskins, & Woodward, 2016 for evidence pointing to relevant cultural variation). All in all, structure-enhancing modifications characteristic of motherese and motionese seem to operate as attention-training techniques that not only cue infants to what is worthy of attention in complex streams of information but also guide their discovery of event structure in ways that optimize processing.

4.4. Developing executive skills promote event-processing fluency

As we have described, information-optimization processes selectively direct attention to information-rich regions within the unfolding sensory flow. Accomplishing such optimization in practice would seem to require the operation of executive processes such as attentional enhancement/suppression and cognitive flexibility to reshape processing in relation to new or unexpected information. These ideas seem to be consistent with others' recent efforts to account for the neural basis of adaptive relations between executive processes and other cognitive processes (e.g., Alexander, Vassena, Deraeve, & Langford, 2017; Vassena, Deraeve, & Alexander, 2019). Considerable evidence points to executive processes undergoing extended development (e.g., Diamond, 2013; Moses & Tahiroglu, 2010; Zelazo, 2015) concomitant with age-related changes in neural circuitry involving development of the prefrontal lobes (e.g., Tsujimoto, 2008). To the extent this is correct

—that event-processing fluency is linked to optimization processes related to developing executive processes—we should expect (a) age-related change in event-processing fluency, and (b) systematic relationships between executive skill and such fluency. Recent findings from our research lab provide initial evidence regarding the second prediction. In particular, using the dwell-time paradigm, we found that preschoolers with weak executive skills displayed both reduced flexibility in reorganizing their attention to an unfolding event relative to children with strong executive skills, and systematically poorer memory for unfolding events (Ross & Baldwin, 2017). We note, however, that while promising, such evidence is correlational, and thus does not yet provide a foundation for a causal claim that executive skills support optimization processes. Indeed, bi-directional relations between the two seem plausible: On the one hand, improving executive skill, such as the ability to selectively enhance or suppress attention, may benefit optimization in event processing, and on the other hand, improvements in optimization and event-processing fluency from other sources (e.g., increasing event knowledge) might in turn benefit executive function. Clearly, further investigation of relations between executive processes and information-optimization processes is needed and warranted.

4.5. Possible links between optimization processes and critical periods for learning

One striking aspect of learning during infancy and childhood is the presence of critical or sensitive periods associated with heightened ability to register complex structural patterning and assemble neural systems that process such patterns efficiently (Lenneberg, 1967; Newport, 2006; Newport, Bavelier, & Neville, 2001; Werker & Hensch, 2015). Whether, and in what way, information-optimization mechanisms might be subject to, and/or play a role in, sensitive period effects is a question of considerable interest. Critical periods occur on varying time tables for different neural systems; recent research begins to reveal neurochemical mechanisms underlying such timing. Of particular interest here is an account of critical periods forwarded relatively recently by Hensch and colleagues (e.g., Hensch, 2005; Werker & Hensch, 2015), in which critical periods are conceptualized as neurochemically induced shifts in the excitatory-inhibitory balance in neural activation. On this account, critical periods are characterized by the up-regulation of exogenously induced neural activation within a given neural system. This openness/receptivity to environmental input is thought to occur as a result of inhibition of endogenously driven activation. During this period, the learner is ideally equipped to register and encode predictability patterns experienced within the sensory flow for the particular neural system at issue, but presumably less able to engage in processing that relies on endogenously driven (e.g., knowledge-based, top-down, executively driven) activation. In the information-optimization account we offer, learners in the midst of a critical window would be advantaged with respect to registering predictability structure as experience streams past, but disadvantaged with respect to their ability to deploy proactive information-optimization sampling of information-rich boundary regions. In contrast, when the critical time window closes, endogenously driven activation would gain ascendancy, with inhibition of environmentally induced neural activation; at this juncture, the learner would

be in a strong position to deploy knowledge-based optimized sampling of the relevant sensory flow, but disadvantaged, relatively speaking, in discovery of new predictability structure within streaming experience. Put another way, critical periods may represent windows of enhanced structure discovery (knowledge accumulation). As critical periods close, endogenously driven, proactive sampling of such discovered structure may predominate, engendering highly efficient apprehension of events within familiar event sequences. Thus, information-optimization processes—specifically, the balance of endogenous (proactive) versus exogenous (reactive) mechanisms within a given neural system—may be the actual locus of critical period effects. If this is correct, critical period effects can be understood within a general, integrative account of event processing more broadly.

4.6. Need for new methods

Specifics of how information optimization might progressively transform processing in infancy and childhood are sparse, largely for two reasons. Until recently, the vast majority of infancy research investigating processing of human action events presented infants with pre-segmented events, thereby rendering moot the possibility of investigating how information-optimization processes underpin detection of event structure within dynamically unfolding sensory streams. As well, sensitive and easily mounted techniques for probing infants' processing of dynamically streaming activity have been scarce. Striking progress on the methodological front is currently being made in a number of research labs. Pupillometry seems to show particular promise in this regard. Pupil dilation in response to perceptual/cognitive engagement is now well-documented (e.g., Goldinger & Pappas, 2012), and recent adult research specifically demonstrates systematic pupil dilation in relation to event boundaries (Tanaka, Kosie, & Baldwin, unpublished manuscript). Findings from pioneering pupillometry research designed to investigate other issues seems also to suggest a comparable pattern of boundary-related pupil dilation in infants as young as 8 months (Jackson & Sirois, 2009). As well, pupillometry seems to offer a strong complement to neuroimaging approaches to investigating event processing, given, as Bilkey and Jensen (2019) eloquently articulate, the inherent interpretive complexity associated with such measures.

In new research, we directly tested the promise that pupillometry holds for indexing infants' sensitivity to structure within novel activity sequences (Kosie, 2019). We found systematic pupil dilation in response to event boundaries in infants only 9–12 months of age, pointing to pupillometry as a valuable index of information-optimization processes at work in infancy. Perhaps even more intriguingly, systematic response to boundaries emerged only when activity unfolded in a highly social “motionese” format, underscoring again our earlier point regarding the significance of social input for promoting infants' registration of the predictability structure of streaming experience. In any case, our new findings validate pupillometry as a method well-suited to investigating a range of key questions about learning more generally, and information optimization, in particular. For example, it may very soon be possible to actually “watch” in real time—via unfolding changes in pupil dilation patterns—as infants first detect predictability structure of various

kinds within streaming experience and implicitly reorganize their sampling of subsequent experience to exploit such structure. Such a methodological advance offers entirely new opportunities to observe processing change as learning is actually underway, rather than simply observing the products of learning, as has been the focus of most prior developmental research.

4.7. *Interim summary*

To sum up quickly on the developmental front, we propose that the events infants are able to apprehend in relation to a given sensory stream change considerably with development, but the fundamental proactive and reactive optimization processes driving event demarcation change considerably less. Current evidence is consistent with infant event processing relying on the self-same information optimization mechanisms that appear to subserve event processing in adults. In other words, we hypothesize that infants, like adults, selectively sample information-rich boundary regions within continuously unfolding activity. On this proposal, what primarily changes about event processing with development is knowledge of predictability structure; as such knowledge is acquired, attentional sampling of the ongoing barrage of dynamically unfolding sensation progressively reorganizes, enabling new event experiences in turn to emerge (see Nagai, 2019 for a related proposal).

What might also change developmentally, in concert with opening and closing of critical periods for learning, is the relative balance of proactive and reactive processes in the registration of unfolding experience. Along the lines suggested by Hensch and colleagues (Hensch, 2016; Werker & Hensch, 2015), critical or sensitive windows may represent periods of enhanced reactive, exogenously driven neural activation that promotes registration of discovered predictability structure, followed by a shift in balance toward enhanced proactive, knowledge-based, endogenously driven neural activation as critical periods wane. In this way, the phenomenon of critical periods in development may be readily integrated into an information-optimization account of event processing.

5. Conclusion

Events are constructed experiences; they are not what actually occurs. What occurs is ongoing dynamic, multidimensional, sensory flow. In question is the nature of the processes that produce event experiences out of the dynamic sensory flow. We propose that the mind transforms the storm of sensory data via implicit mechanisms of information optimization—both proactive and reactive—into structured, potentially describable, memorable units of experience. Our domain-general information-optimization account extends Zacks and colleagues' EST in important ways. In particular, we have shown that, unlike EST, the information-optimization perspective accounts for viewers' proactive sampling of predictably unpredictable boundary content within dynamically unfolding sensory streams. As well, we articulated how information optimization operates in the context of

novelty to guide learning. In particular, viewers' reactive response to novelty and prediction failure provides opportunity for discovery of new predictability structure, which progressively reshapes processing over time as such newly discovered structure is incorporated to guide processing proactively. Our recent studies (Hard et al., 2019; Kosie & Baldwin, 2019) showcase such reorganization as it unfolds in time. In this way, the information-optimization framework readily explains developmental change by (a) locating change to a key degree in the way that increasing familiarity with predictability structure (knowledge) translates to reorganized deployment of attention, with these changes potentiated by information-optimization processes operative from earliest infancy, and (b) accounting for critical or sensitive periods as shifts in the balance of proactive versus reactive mechanisms of information optimization, with reactive mechanisms privileged as critical windows open, and proactive mechanisms guided by thus-far-acquired knowledge of predictability structure gaining ascendancy as critical periods close.

Although in some respects our information-optimization account is speculative, there is nevertheless considerable confirmatory evidence already available. Our review identified a range of fruitful avenues for continuing investigation. The advent of non-invasive methods for probing cognitive engagement such as dwell time and pupillometry, which are well-suited to investigating processing of streaming information, heralds an era of potentially explosive progress regarding these fundamental questions lying at the heart of the science of learning and development.

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